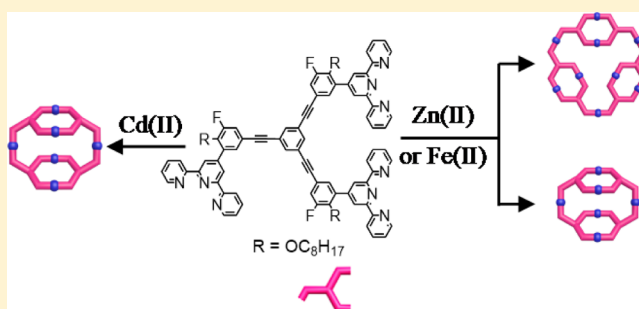


Self-Assembly of Tetrameric and Hexameric Terpyridine-Based Macrocycles Using Cd(II), Zn(II), and Fe(II)

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S Supporting Information

ABSTRACT: The self-assembly behavior of a tritopic 2,2':6',2''-terpyridine (tpy) ligand with Cd(II), Zn(II), and Fe(II) has been exploited herein to generate a series of tetrameric and hexameric macrocycles. The main advantage of using such transition metals with an octahedral coordination geometry is their distinct coordination abilities (e.g., binding strength and reversibility). With the same ligand, this study reveals that the supramolecular structural variation between tetrameric and hexameric macrocycle architectures can be precisely controlled using different metal ions with the same coordination geometry. When Cd(II) was used, a tetrameric macrocycle was the only observed structure in the self-assembly, whereas Zn(II) and Fe(II) assembled a mixture of tetrameric and hexameric macrocycles. Because of the high stability of Fe(II) as the coordination center, we successfully isolated tetrameric and hexameric macrocycles using a regular column. In-depth characterization was carried out to establish the proposed structures, including multinuclear NMR (¹H, ¹⁹F, and ¹³C) analysis, electrospray ionization mass spectrometry, and 2D ion-mobility mass spectrometry.



■ INTRODUCTION

The precision with which biology forms near monodisperse proteins or enzymes from peptides has inspired chemists to design and create complex supramolecules using self-assembly as an elegant and energy-efficient bottom-up approach.¹ In recent years, coordination-driven self-assembly has made considerable progress in constructing a myriad of supramolecular architectures, including one-dimensional (1D) helicates, two-dimensional (2D) macrocycles, knots, links, three-dimensional (3D) cages, and capsules.^{2–12} This active research interest in the field has been maintained and accelerated by a large variety of applications such as catalysis,^{5b,13} host–guest chemistry,¹⁴ light-harvesting and light-emitting devices,¹⁵ gas storage,^{11,16} and chemical separation and sensing.¹⁷

The generation of such functional supramolecular architectures requires precise control over self-assembly and a deep understanding of the factors that determine the size and shape of the assemblies. The current study shows that small variations in the ligands' geometries,^{4a,18} metal ions,¹⁹ anion,^{6b,20} time,²¹ concentration,²² and solvent^{12b,23} often result in large variations in the structure and topology. In a truly pioneering work, Lehn and co-workers reported that the same tris-bipyridine ligand

yielded a 1D triple helix^{19a} and a 2D pentanuclear circular double helix^{19b} with Ni(II) and Fe(II), respectively. Among these two different structures, the 2D circular double helix was demonstrated as a thermodynamic product, while the 1D triple helix was suggested as a kinetic product trapped in local minima.²¹ Furthermore, self-assembly of the same tris-bipyridine ligand with Fe(II) salts formed 2D penta- or hexanuclear circular helicates depending on the counterions. After that, numerous studies reported substantial changes in the structure of the final assemblies of 2D²⁰ and 3D^{6b,c,16b,19c,24} systems based on the counterions and metal ions.

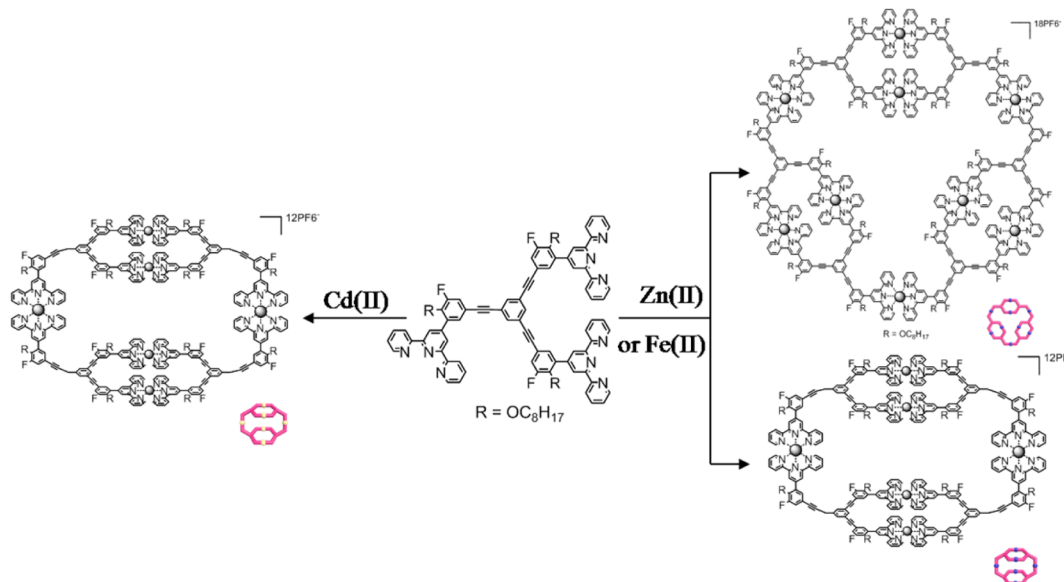
In a 2,2':6',2''-terpyridine (tpy)-based supramolecular chemistry field,²⁵ an impressive number of complexes have been prepared by varying the nature of the metal and by introducing various substituents onto the tpy part, including supramolecular polymers,^{15b,26} macrocycles,²⁷ cages,²⁸ and 2D complex architectures.²⁹ Although tpy has attracted widespread attention because of its excellent complexing properties as N-donor ligands toward numerous main-group, transition-metal,

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Scheme 1. Self-Assembly of Tetrameric and Hexameric Macrocycles Using **L** with Cd(II), Zn(II), and Fe(II) Ions

and lanthanide cations, using the same ligands with different metal ions only led to the assembly of analogous supra-molecular structures. In principle, we might be able to precisely tune the self-assembly into various architectures by using different metal ions. Therefore, it was of interest to study the features controlling the self-assembly of different architectures, in particular, how variation of the metal ions would affect the nature of the assembly generated without changing the ligand.

We report here that the self-assembly of a tritopic tpy ligand with different metal ions with octahedral coordination geometry, viz., Cd(II), Zn(II), and Fe(II) salts, leads to the successful production of tetrameric and hexameric macrocycles (Scheme 1). Those two macrocycles display different molecular geometries and symmetries according to molecular modeling (Figure 1). The main advantage of using those transition metals

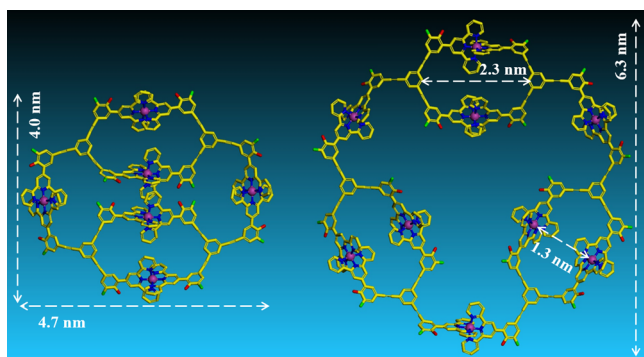


Figure 1. Energy-minimized structures of tetrameric and hexameric macrocycles from molecular modeling (the alkyl chains are omitted for clarity).

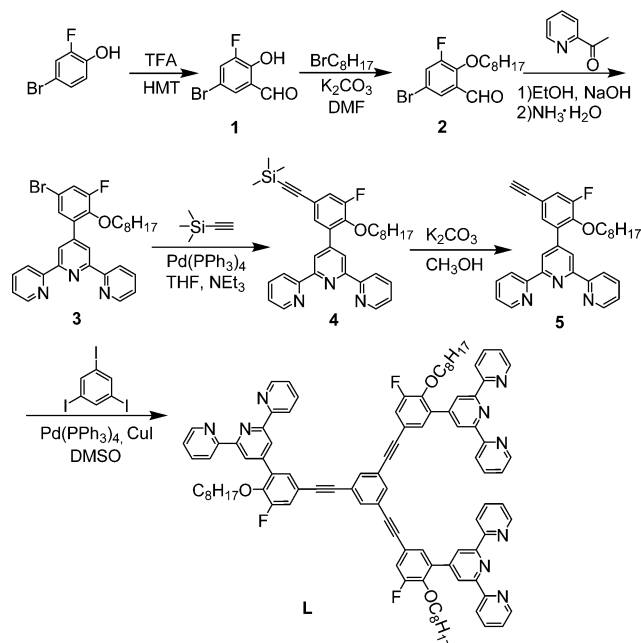
with tpy is the significantly different stability of the metal complex: Cd(II) < Zn(II) < Fe(II).³⁰ When Cd(II) was used, a tetrameric macrocycle was the observed structure in the self-assembly, whereas Zn(II) and Fe(II) assembled a mixture of tetrameric and hexameric macrocycles. Because of the high stability of Fe(II) as a coordination center, we successfully

isolated tetrameric and hexameric macrocycles using a regular column.

RESULTS AND DISCUSSION

Ligand Synthesis. The tritopic tpy ligand **L** was synthesized by Sonogashira coupling of 1,3,5-triodobenzene with the alkynylterpyridine **5** (Scheme 2). Similar to our

Scheme 2. Synthesis of Ligand **L**



previous study,^{29c} compound **3** was synthesized by reacting 5-bromo-3-fluoro-2-(octyloxy)benzaldehyde with 2-acetylpyridine in a one-pot process, which included the formation of diketone intermediate species and cyclization by $\text{NH}_3 \cdot \text{H}_2\text{O}$. Alkyl chains were incorporated in order to increase the solubility and simplify the aromatic region of the ^1H NMR spectrum of **L**. Particularly, fluorine was also introduced to the backbone of **L** to facilitate multinuclear NMR characterization

(^1H , ^{19}F , and ^{13}C) of supramolecular complexes. The ^1H and ^{19}F NMR spectra of **L** showed a single set of signals corresponding to the symmetric structure (Figure 3). Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS) also showed a single peak for **L** (Figure S17).

Self-Assembly of a Cd(II) Complex. Because of the high lability and reversibility of $\langle \text{tpy}-\text{Cd(II)}-\text{tpy} \rangle$, we initially combined **L** and $\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ with a precise stoichiometric ratio of 2:3 for self-assembly at 50°C for 8 h, followed by the addition of excess NH_4PF_6 , to give a white precipitate with 91% isolated yield. Conventional electrospray ionization mass spectrometry (ESI-MS) provided readily assignable peaks corresponding to the composition of Cd_6L_4 (molecular weight, 8461.8 Da) with continuous charge states (5+ to 10+) by losing different numbers of PF_6^- counterions (Figure 2A). The

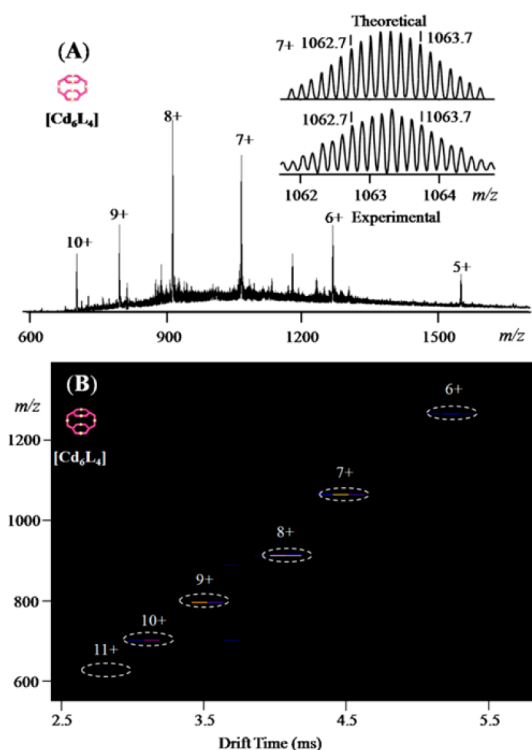


Figure 2. (A) ESI-MS and (B) 2D IM-MS spectra (m/z vs drift time) of complex Cd_6L_4 (2.0 mg/mL). The charge states of intact assemblies are marked.

experimental isotope patterns agreed well with the simulated peaks. Ion-mobility mass spectrometry (IM-MS),^{27a,31} which is a powerful tool to differentiate isomers or conformers based on their different collision cross sections, displayed a series of narrow signals at each charge state, suggesting the existence of a tetramer as the major assembled structure (Figure 2B).

In the ^1H NMR spectrum of Cd_6L_4 , we observed two sets of tpy signals, with the 2:1 ratio indicating the lower symmetric structure (Figure 3A). The 2D COSY NMR spectrum of Cd_6L_4 also showed two sets of signals (Figure S21). The shift of the aromatic protons from tpy units and phenyl groups was similar to that of previous reports.^{29c} Other than the protons at the 6 position of tpy dramatically shifted upfield ($\Delta\delta = 0.6$ ppm) due to the electron shielding effect after complexation with metal ions, all other aromatic proton peaks of tpy and phenyl were shifted slightly downfield. A proposed tetrameric macrocycle

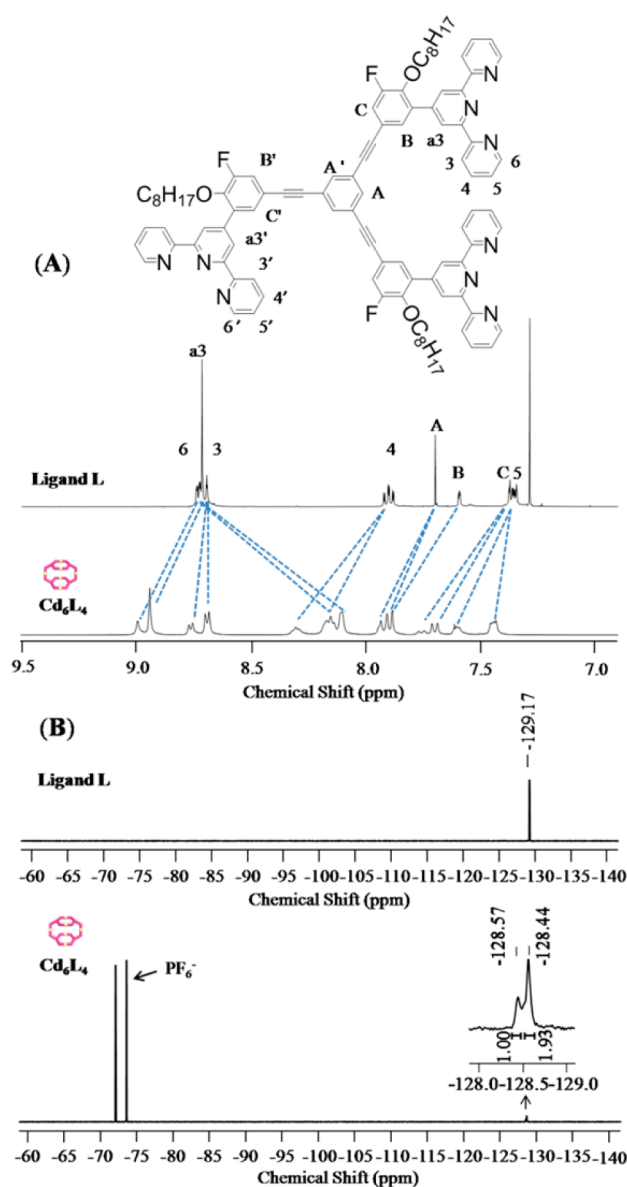


Figure 3. (A) ^1H NMR spectra (500 MHz) of ligand **L** in CDCl_3 and complex Cd_6L_4 in CD_3CN (5.0 mg/mL). (B) ^{19}F NMR spectra of ligand **L** in CDCl_3 and complex Cd_6L_4 in CD_3CN .

with Cd_6L_4 composition was depicted in Scheme 1 according to a previous study.^{29h,i} In the ^{19}F NMR spectrum, the two peaks around 72 ppm were assigned to PF_6^- .³² Meanwhile, a clear diagnostic feature was the presence of two different signals at -128.57 and -128.44 ppm, suggesting the formation of architecture with two types of tpy units (Figure 3B). Accordingly, pure **L** only showed a single peak at -129.17 ppm.

We attribute the formation of the tetrameric macrocycle to the stable small irregular hexagon (Figure 4), which may act as an intermediate before reaching the final structure. To validate our hypothesis, **L** and $\text{Cd}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ were mixed with a precise stoichiometric ratio of 1:1 for self-assembly. As expected, we observed the formation of a hexagon-like dimer in ESI-MS as the predominant structure (Figure 4). This inductive effect through the formation of a hexagon-like dimer was also observed in our previous study with a different tritopic tpy ligand.^{29c} Additionally, Cd(II) with a larger ionic radius, bearing greater flexibility, allowed for a considerable deviation

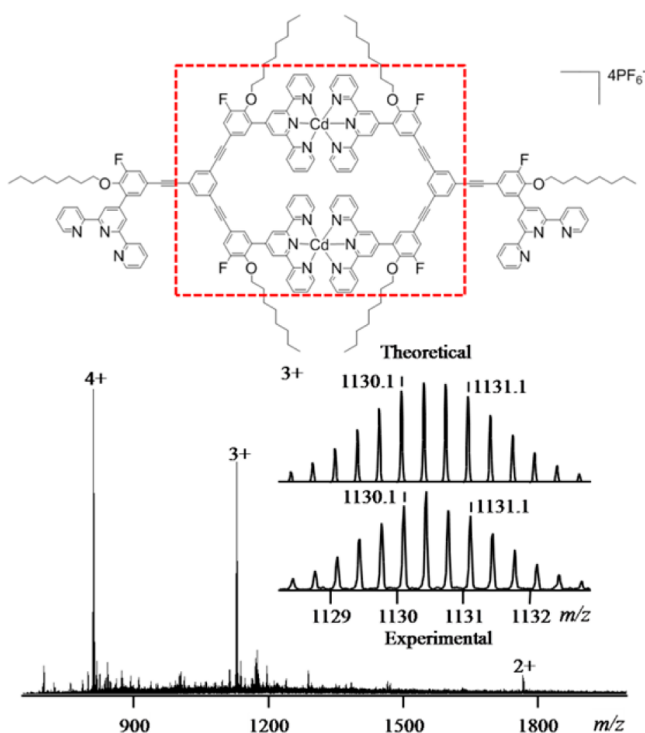


Figure 4. ESI-MS spectrum and isotope pattern of the dimer obtained by the self-assembly of ligand **L** with Cd(II) at a molar ratio of 1:1.

in its coordination geometry to form a slightly bent tetrameric macrocycle, as we proposed with a stoichiometric ratio of 2:3.

More interestingly, Zhang and co-workers recently reported a similar organic tetrameric structure that contained two macrocyclic panels and exhibited D_{2h} symmetry through dynamic covalent assembly.³³ Similarly, the formation of a small irregular hexagon was proven to be the key intermediate in the assembly of such a pure organic tetramer. Therefore, the assembly of such bent tetrameric macrocycles in a dynamic noncovalent approach coincides perfectly with the behavior of a dynamic covalent approach. This coincidence broadens our understanding of the structure design and self-assembly.

Self-Assembly of a Zn(II) Complex. These initial results motivated us to perform the self-assembly using Zn(II), which has a slightly stronger binding strength with tpy than Cd(II).³⁰ In contrast to the single product observed using Cd(II), the self-assembly of Zn(II) produced two structures according to ESI-MS detection, i.e., tetramer and hexamer with **L** (Figure 5A). It is worth noting that IM-MS clearly separated the differently charged ions of tetramer and hexamer at the same m/z , e.g., Zn_4L_4 versus Zn_6L_6 , which were overlapped at some charge states in conventional ESI-MS (Figure 5B). However, we were not able to isolate these two assemblies from the mixture because the weak connectivity of $\langle tpy-Zn(II)-tpy \rangle$ was not stable enough to sustain regular chromatographic separation.

Nevertheless, only two sets of signals were detected in ^{19}F and 1H NMR spectra, which were similar to the spectra of the Cd(II) tetramer (Figures S24 and S25). We inferred that the Zn(II) tetramer and hexamer could have NMR patterns similar to those of other tpy-based macrocycles reported.^{27b,34} However, 2D COSY NMR showed a split correlation between two characteristic protons (i.e., the 4 position with the 5 position) on tpy (Figure S27), indicating that two species

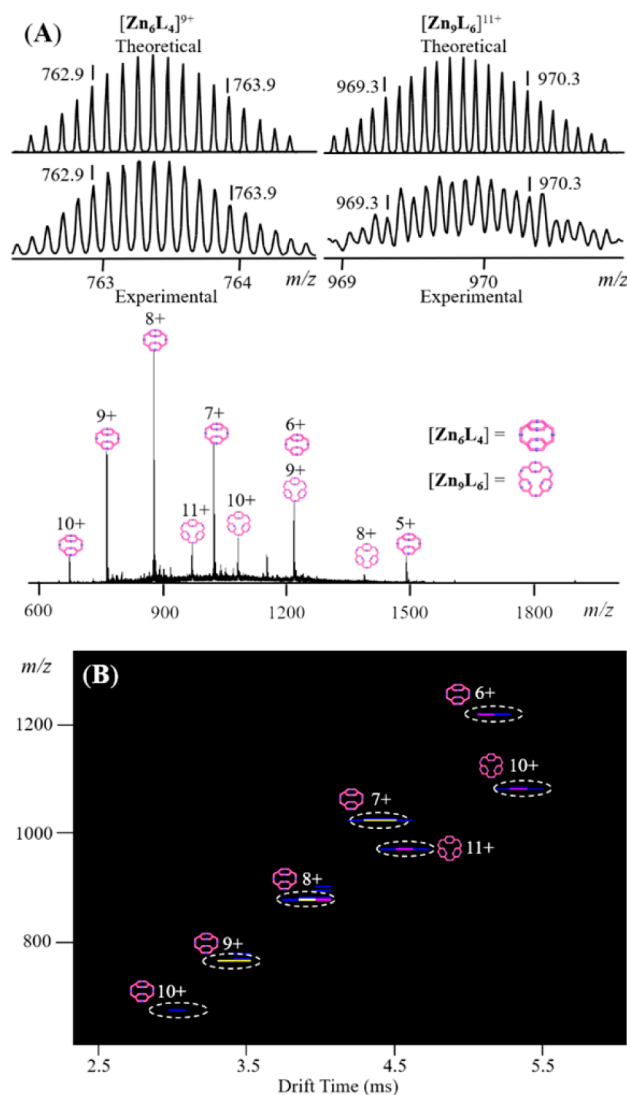


Figure 5. (A) ESI-MS and (B) 2D IM-MS spectra (m/z vs drift time) of complexes Zn_4L_4 and Zn_6L_6 (2.0 mg/mL). The charge states of intact assemblies are marked.

existed. Therefore, we propose tetrameric and hexameric macrocycles for the Zn(II) self-assembly, as shown in Scheme 1. The next challenge was isolating and characterizing these two complexes.

Self-Assembly and Separation of a Fe(II) Complex. As mentioned earlier, the distinct coordination ability (e.g., binding strength and reversibility) of transition metals provides us with multiple options for self-assembly. The strong connectivity of $\langle tpy-Fe(II)-tpy \rangle$ is stable enough to sustain a regular chromatographic separation.^{29fh} With $FeSO_4 \cdot 7H_2O$ as the metal source, we followed the same self-assembly procedure using trichloromethane/methanol ($CHCl_3/MeOH$) as the solvent at 50 °C. However, we only observed many polymeric products in ESI-MS. Then ethylene glycol was used as the solvent for self-assembly at 160 °C for 3 days under N_2 protection. We successfully separated both tetramer Fe_4L_4 and hexamer Fe_6L_6 using a regular silica column with isolation yields of 23% and 50%, respectively. This indicates that the high temperature significantly improves the reversibility of the $\langle tpy-Fe(II)-tpy \rangle$ connectivity and prevents the formation of random polymeric assemblies.

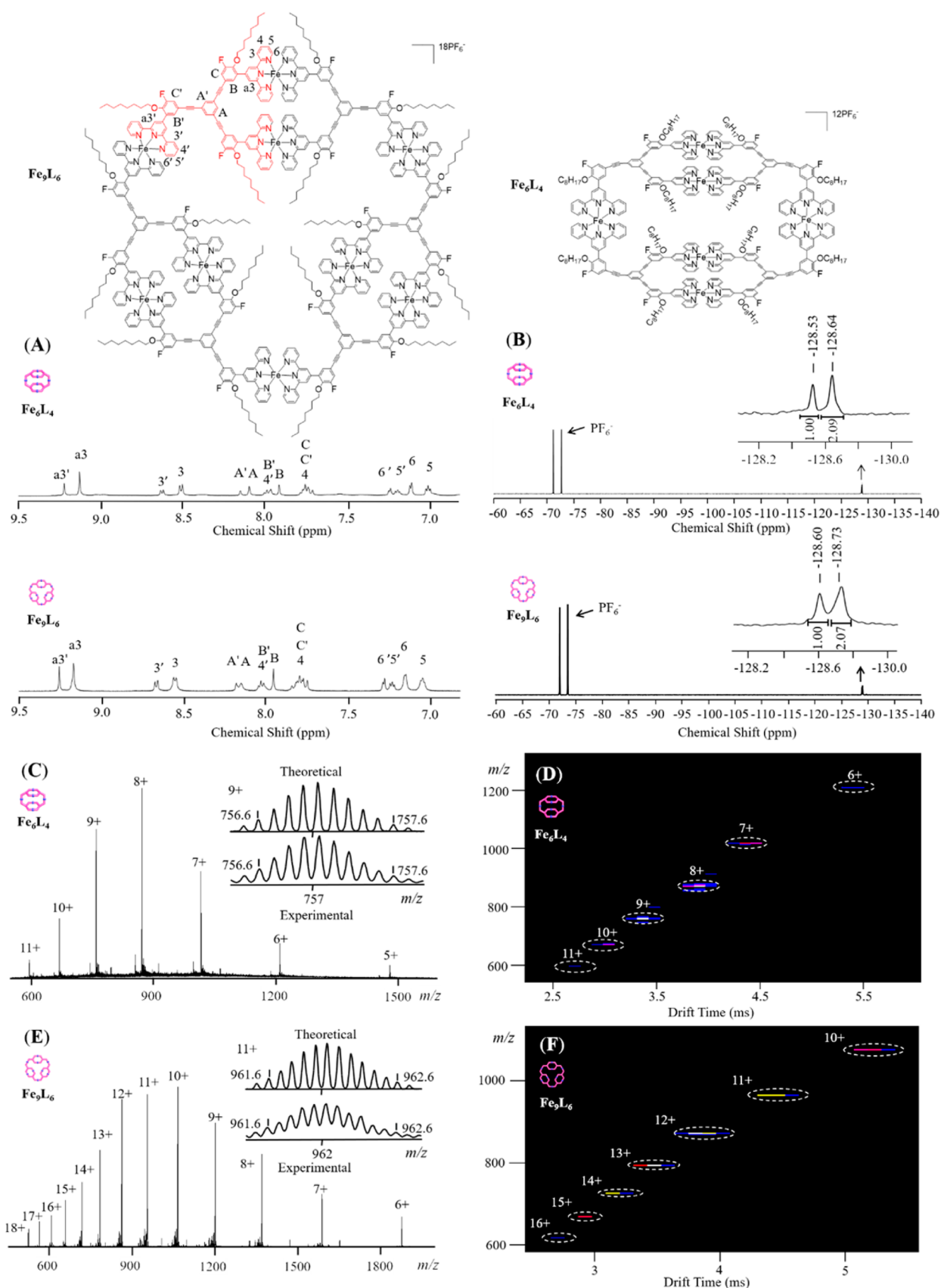


Figure 6. (A) ^1H NMR spectra of complexes Fe_6L_4 and Fe_9L_6 in CD_3CN (5.0 mg/mL). (B) ^{19}F NMR spectra of Fe_6L_4 and Fe_9L_6 in CD_3CN . (C) ESI-MS and (D) 2D IM-MS spectra (m/z vs drift time) of Fe_6L_4 (2.0 mg/mL). (E) ESI-MS and (F) 2D IM-MS spectra (m/z vs drift time) of Fe_9L_6 (2.0 mg/mL). The charge states of intact assemblies are marked.

ESI-MS and IM-MS suggested their tetrameric and hexameric compositions (Figure 6C–F). The most conclusive evidence indicating their 2D macrocyclic structures was provided by ^1H and ^{19}F NMR, which unambiguously exhibited two sets of signals corresponding to the proposed structures. As expected, the ^1H and ^{19}F NMR spectra of tetramer and hexamer exhibited two sets of tpy signals with almost identical distribution owing to their cyclic geometry (Figure 6A,B), as we predicted in the self-assembly with Zn(II). For instance, ^{19}F NMR spectra displayed two broad peaks at around -128.61 and -128.70 ppm for both tetrameric and hexameric macrocycles (Figure S25). Compared to the similar NMR pattern and mass spectrometry results observed for the Zn(II) self-assembly, we reason that Zn(II) and Fe(II) should have similar self-assembly behavior except for their different stabilities. Very recently, Newkome and co-workers reported the separation of tetramer using the combination of Ru(II) and Fe(II) with a three-armed tpy ligand on a triphenylamine scaffold; however, no hexamer was reported in their study.^{29h} We speculated that triphenylamine may introduce more flexibility to accommodate the ring strain in the formation of a tetramer compared to our 1,3,5-triethynylbenzene core. As such, tetramer was the predominant structure in their case in an entropy-driven self-assembly.

To further confirm this conclusion, we also performed a control study similar to that of Cd(II) (Figure 4) by mixing L and Zn(II) or Fe(II) with a precise stoichiometric ratio of 1:1 for self-assembly. As expected, we also observed the formation of hexagon-like dimers in ESI-MS as the preferable structure, which should serve as the intermediates of the final assemblies (Figures S7 and S8). Moreover, the outcome of this comparison may advance our understanding to the supramolecular characterization by NMR, in which single sets of signals do not always correspond to single discrete structures, particularly for 2D macrocycles. Taken together, we speculated that the formation of Cd(II) tetramer as the sole product was attributed to the large size of Cd(II) with considerable coordination deviation.

Further Characterization by IM-MS and 2D DOSY NMR. We further compared the size difference of Cd(II), Zn(II), and Fe(II) tetramers assembled by L using IM-MS drift times (Table 1), which can be assumed as the identity of each

Table 1. Experimental Drift Times (ms) of Tetramers

	Cd_6L_4	Zn_6L_4	Fe_6L_4
6+	5.07	4.96	5.18
7+	4.30	4.19	4.19
8+	3.86	3.75	3.75
9+	3.31	3.20	3.20
10+	2.98	2.87	2.87
11+	2.65		2.54

structure. In ion mobility, compact structures always travel faster than more elongated (extended) ions at the same charge because of fewer interactions with the buffer gas in the ion-mobility chamber.^{27a,31} The tetramers of Zn(II) and Fe(II) were recorded with the same drift times at different charge states except for 6+, suggesting the same assembled 2D macrocyclic structures due to the similar ionic radii of metal ions, i.e., 88 pm for Zn(II) and 75 pm for Fe(II). The Cd(II) tetrameric macrocycle, however, exhibited a slightly large drift time because of a large radius at 109 pm. Diffusion-ordered

NMR spectroscopy (DOSY) data are shown in Figure 7. In contrast to the other three supramolecules Cd_6L_4 , Fe_6L_4 , and

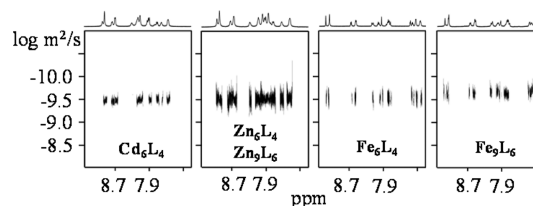


Figure 7. 2D DOSY NMR spectra of these four supramolecular assemblies (500 Hz, 5.0 mg/mL in CD_3CN , 300 K).

Fe_9L_6 , the DOSY spectrum of the self-assembly by L and Zn(II) displays two sets of signals corresponding to the mixture of tetrameric and hexameric macrocycles.

Cyclic Voltammetry. The electrochemical properties of Fe_6L_4 and Fe_9L_6 in *N,N*-dimethylformamide (DMF) were studied in a three-electrode electrochemical cell with Bu_4NPF_6 (0.001 M) as the electrolyte. A platinum button working electrode was used with a platinum flag counter electrode. A silver wire pseudoreference electrode was calibrated using the ferrocene/ferricinium couple³⁵ (Figure 8). Both complexes

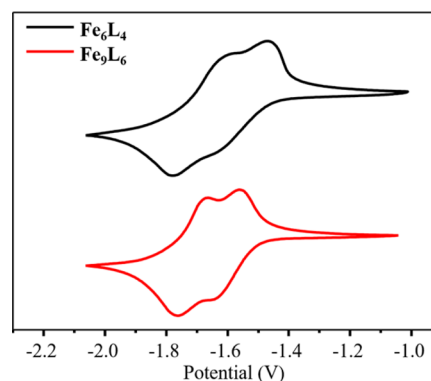


Figure 8. Cyclic voltammograms of complexes Fe_6L_4 and Fe_9L_6 (0.001 M in a 0.1 M solution of Bu_4NPF_6 in DMF) with a scan rate at 100 mV/s.

exhibited two tpy ligand-centered redox couples in the negative potential region, indicating that there were two kinds of electron-transfer environments for Fe(II) in each complex.^{36,37} Meanwhile, the redox potentials of two complexes appeared at very close positions ($E^1_{1/2} = -1.70$ V and $E^2_{1/2} = -1.54$ V for Fe_6L_4 and $E^1_{1/2} = -1.72$ V and $E^2_{1/2} = -1.59$ V for Fe_9L_6), suggesting similar redox properties for the two macrocycles. However, under a zoomed-in comparison, Fe_9L_6 showed a slightly smaller peak-to-peak separation ($\Delta E_p = 60$ mV) for the redox couple at a lower (less negative) potential region than Fe_6L_4 ($\Delta E_p = 110$ mV). This indicated that Fe_9L_6 had a higher reversibility and a faster electron-transfer rate than Fe_6L_4 under the same conditions because of an increase of the localized repeating units and a decrease of the band gap.³⁸

Fabrication of Nanostructures. Considering the long alkyl chains of those macrocycles, we investigated their hierarchical self-assembly behavior in different solvents. It was found that all of the macrocycles (Cd_6L_4 , Fe_6L_4 , and Fe_9L_6) could form spherical nanostructures in the mixed solvents of DMF and tetrahydrofuran (THF) (1:4, v/v). Transmission electron microscopy (TEM) measurements were performed by

depositing the solution of macrocycles on copper grids, followed by drying under vacuum at 40 °C. Spherical aggregates with diameters of 50–300 nm were observed (Figure 9). Those aggregates were formed through multiple intermolecular interactions (e.g., π – π stacking, CH– π interactions) according to a previous report.^{39,40}

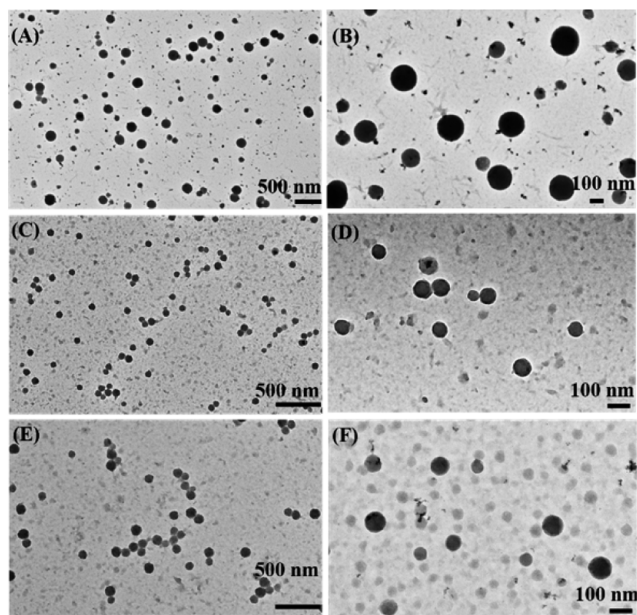


Figure 9. TEM images of spherical aggregates of Cd₆L₄ (A and B), Fe₆L₄ (C and D), and Fe₉L₆ (E and F).

CONCLUSION

The comprehensive investigation reported herein represents the self-assembly behavior of the same tristerpyridine ligands with Cd(II), Zn(II), and Fe(II) based on their different coordination abilities, e.g., binding strength and reversibility. When Cd(II) was used as the coordination center, the self-assembly favored the formation of tetrameric macrocycles perhaps because of the large size of Cd(II) with considerable coordination deviation. In the case of Zn(II) and Fe(II), a mixture of tetrameric and hexameric macrocycles was obtained. The tetrameric and hexameric macrocycles with Fe(II) were successfully isolated and displayed identical NMR spectra (¹H and ¹⁹F), reminding us of the limitation of NMR characterization for macrocycles. The evidence of forming intermediate dimers for Cd(II), Zn(II), and Fe(II) suggested their similar self-assembly behavior. In addition to the previous reports on the structural variation between 1D and 2D helices, our case on the structural variation between 2D macrocycles will enrich the library of supramolecular structures and enhance our fundamental understanding of the supramolecular assembly.

EXPERIMENTAL SECTION

Sonogashira coupling reactions were carried out using standard Schlenk-line techniques under an inert atmosphere of nitrogen. ¹H and ¹³C NMR spectral data were recorded on a Bruker Avance 400 MHz/500 MHz NMR spectrometer and a Varian Inova 400 MHz spectrometer in CDCl₃ and CD₃CN with a tetramethylsilane standard. ¹⁹F NMR spectral data for organic compounds were recorded on a Varian Inova 400 MHz spectrometer in CDCl₃ (trichlorofluoromethane was added as the reference, 0 ppm). ¹⁹F NMR spectral data for the complexes were recorded on a Bruker Avance 500 MHz

spectrometer in CD₃CN (PF₆[−] was viewed as the reference).³² MALDI-TOF MS spectra were recorded with a Bruker Auto Flex II mass spectrometer using *trans*-2-[3-(4-*tert*-butylphenyl)-2-methyl-2-propenylidene]malononitrile (DCTB) as the matrix. ESI-MS was conducted on a Waters Synapt G2 mass spectrometer with traveling-wave ion mobility.

Synthesis of Ligand L. Under nitrogen protection, a mixture of Pd(PPh₃)₄ (52 mg, 0.05 mmol), CuI (7.6 mg, 0.04 mmol), 1,3,5-triiodobenzene (407 mg, 0.9 mmol), and compound 5 (1.5 g, 3.1 mmol) in 25 mL of dimethyl sulfoxide (DMSO) and 15 mL of triethylamine was stirred at 80 °C for 48 h. After cooling to room temperature, the residue was extracted with dichloromethane and washed three times using water. The solvent was removed, and the residue was purified by column chromatography on SiO₂ with chloroform (adding 1% ethanol) as the eluent to afford L in 72% yield as a white solid. ¹H NMR (500 MHz, CDCl₃): δ 8.76–8.73 (m, 6H, tpy H⁶), 8.73–8.70 (m, 12H, tpy H³ and tpy H^{3'}), 7.91 (td, J = 9.8 and 2.3 Hz, 6H, tpy H⁴), 7.69 (s, 3H, Ph H^A), 7.59 (dd, J = 2.5 and 1.6 Hz, 3H, Ph H^B), 7.39–7.33 (m, 9H, tpy H⁵ and Ph H^C), 4.03 (td, J = 8.0 and 1.0 Hz, 6H, alkyl H^D), 1.65–1.57 (m, 6H, alkyl H^E), 1.27–1.18 (m, 6H, alkyl H^F), 1.18–1.12 (m, 6H, alkyl H^G), 1.11–0.98 (m, 18H, alkyl H^G, alkyl H^I and alkyl H^J), 0.80 (t, J = 9.0 Hz, 9H, alkyl H^K). ¹³C NMR (125 MHz, CDCl₃): δ 156.61, 156.12, 155.46, 154.15, 149.19, 146.54, 146.51, 145.51, 145.40, 136.74, 134.77, 134.74, 134.19, 129.34, 129.31, 123.81, 123.76, 121.49, 121.17, 120.17, 119.96, 118.18, 118.08, 89.14, 89.11, 88.18, 74.76, 74.70, 31.71, 29.94, 29.23, 29.03, 25.77, 22.55, 14.02. ¹⁹F NMR (376 MHz, CDCl₃): δ −129.17. MALDI-TOF MS. Calcd for [C₉₉H₉₀F₃N₉O₃ + H]⁺: m/z 1510.7. Found: m/z 1510.6.

Complex Cd₆L₄. To a solution of ligand L (7.2 mg, 4.8 μ mol) in CHCl₃ (1.0 mL) was added a solution of Cd(NO₃)₂·4H₂O (2.2 mg, 7.2 μ mol) in MeOH (3 mL). The mixture was stirred at 50 °C for 8 h. After cooling to room temperature, 200 mg of NH₄PF₆ was added to bring the product out of the solution. The precipitate was washed using water by centrifugation and dried under vacuum (yield 91%). ¹H NMR (500 MHz, CD₃CN): δ 8.99 (s, 2H, tpy H^{3'}), 8.94 (s, 4H, tpy H³), 8.76 (d, J = 8.1 Hz, 2H, tpy H^{3'}), 8.69 (d, J = 8.2 Hz, 4H, tpy H³), 8.30 (d, J = 6.9 Hz, 2H, tpy H^{4'}), 8.20–8.12 (m, 6H, tpy H⁴ and tpy H^{6'}), 8.10 (s, 4H, tpy H⁶), 7.93 (s, 2H, Ph H^{A'}), 7.91 (s, 2H, Ph H^A and Ph H^{B'}), 7.88 (s, 2H, Ph H^B), 7.76 (d, J = 11.4 Hz, 1H, Ph H^{C'}), 7.70 (d, J = 12.0 Hz, 2H, Ph H^C), 7.63–7.58 (m, 2H, tpy H^{5'}), 7.47–7.41 (m, 4H, tpy H⁵), 4.26 (m, 6H, alkyl H^D and alkyl H^{D'}), 1.75–1.64 (m, 6H, alkyl H^E and alkyl H^{E'}), 1.30 (m, 6H, alkyl H^F and alkyl H^{F'}), 1.14 (m, 6H, alkyl H^G and alkyl H^{G'}), 0.91 (m, 18H, alkyl H^G, alkyl H^{I'}, alkyl H^I, alkyl H^{I'}, alkyl H^J, and alkyl H^{J'}), 0.66 (m, 9H, alkyl H^K and alkyl H^{K'}). ¹³C NMR (125 MHz, CD₃CN): δ 156.30, 156.27, 154.33, 154.30, 151.68, 151.64, 149.78, 149.73, 149.70, 149.62, 148.69, 148.61, 145.75, 145.71, 145.65, 145.62, 141.60, 141.46, 133.90, 132.57, 129.87, 129.47, 127.55, 127.36, 124.70, 124.58, 124.02, 123.72, 123.69, 118.30, 118.22, 88.91, 88.27, 75.19, 75.14, 31.46, 31.43, 29.99, 29.94, 29.16, 29.09, 29.04, 29.01, 26.11, 25.98, 22.26, 22.21, 13.28, 13.24. ESI-MS for Cd₆L₄ (m/z): 1969.5 ([M − 4PF₆]⁴⁺; calcd m/z 1969.5), 1546.5 ([M − 5PF₆]⁵⁺; calcd m/z 1546.5), 1264.7 ([M − 6PF₆]⁶⁺; calcd m/z 1264.7), 1063.3 ([M − 7PF₆]⁷⁺; calcd m/z 1063.3), 912.3 ([M − 8PF₆]⁸⁺; calcd m/z 912.3), 794.7 ([M − 9PF₆]⁹⁺; calcd m/z 794.7), 700.8 ([M − 10PF₆]¹⁰⁺; calcd m/z 700.8).

Complexes Zn₆L₄ and Zn₉L₆. To a solution of ligand L (7.0 mg, 4.6 μ mol) in CHCl₃ (1.0 mL) was added a solution of Zn(NO₃)₂·6H₂O (2.1 mg, 7.0 μ mol) in MeOH (3 mL); then the mixture was stirred at 50 °C for 8 h. After cooling to room temperature, 200 mg of NH₄PF₆ was added, and a yellow precipitate was observed. The precipitate was washed using water by centrifugation and dried under vacuum (yield 83%). ESI-MS for Zn₆L₄ (m/z): 1548.3 ([M − 4PF₆]⁴⁺; calcd m/z 1548.3), 1489.9 ([M − 5PF₆]⁵⁺; calcd m/z 1489.9), 1217.5 ([M − 6PF₆]⁶⁺; calcd m/z 1217.5), 1022.8 ([M − 7PF₆]⁷⁺; calcd m/z 1022.8), 876.9 ([M − 8PF₆]⁸⁺; calcd m/z 876.9), 763.4 ([M − 9PF₆]⁹⁺; calcd m/z 763.4), 672.5 ([M − 10PF₆]¹⁰⁺; calcd m/z 672.5), 598.3 ([M − 11PF₆]¹¹⁺; calcd m/z 598.3). ESI-MS for Zn₉L₆ (m/z): 1387.9 ([M − 8PF₆]⁸⁺; calcd m/z

1387.9), 1081.2 ($[M - 10PF_6]^{-10+}$; calcd m/z 1081.2), 969.8 ($[M - 11PF_6]^{-11+}$; calcd m/z 969.8).

Complexes Fe_6L_4 and Fe_9L_6 . Ligand **L** (56 mg, 37.1 μ mol) and $FeSO_4 \cdot 7H_2O$ (15.5 mg, 55.7 μ mol) were dissolved in 40 mL of ethylene glycol. The solution was heated at 160 °C for 3 days under N_2 protection. After that, the solution was cooled and added into a 60 mL MeOH solution with 2 g of NH_4PF_6 . This solution was centrifuged, and the residue was flash-column-chromatographed (SiO_2), eluting with MeCN/saturated $KNO_3(aq)$ (100:0–100:15). After removal of the volatile, the mixture was filtered and washed with water. The solid was dissolved in MeOH (4 mL) and dropped into a solution with NH_4PF_6 (240 mg in 8 mL of MeOH). The purple precipitate was washed with water, and the product was obtained as a solid.

Fe_6L_4 . Yield: 23%. 1H NMR (500 MHz, CD_3CN): δ 9.20 (s, 2H, tpy $H^{3'}$), 9.17 (s, 4H, tpy $H^{3'}$), 8.62 (d, $J = 8.1$ Hz, 2H, tpy $H^{3'}$), 8.51 (d, $J = 8.0$ Hz, 4H, tpy $H^{3'}$), 8.14 (s, 1H, Ph H^A), 8.09 (s, 2H, Ph H^A), 8.01–7.94 (m, 3H, tpy $H^{4'}$ and Ph H^B), 7.91 (d, $J = 1.3$ Hz, 2H, Ph H^B), 7.73 (m, 7H, tpy H^4 , Ph $H^{C'}$, and Ph H^C), 7.23 (d, $J = 5.4$ Hz, 2H, tpy $H^{6'}$), 7.21–7.15 (m, 2H, tpy $H^{5'}$), 7.10 (d, $J = 5.5$ Hz, 4H, tpy H^6), 6.99 (t, $J = 6.6$ Hz, 2H, tpy H^5), 4.44 (t, $J = 5.7$ Hz, 2H, alkyl $H^{D'}$), 4.33 (t, $J = 5.6$ Hz, 4H, alkyl H^D), 1.90–1.84 (m, 2H, alkyl $H^{E'}$), 1.79–1.71 (m, 4H, alkyl H^E), 1.50 (m, 2H, alkyl $H^{F'}$), 1.38 (m, 4H, alkyl H^F), 1.29–1.21 (m, 2H, alkyl $H^{G'}$), 1.12 (m, 4H, alkyl H^G), 0.96 (m, 6H, alkyl $H^{H'}$, alkyl H^I , and alkyl $H^{J'}$), 0.82 (m, 12H, alkyl $H^{H'}$, alkyl H^I , and alkyl H^J), 0.62 (t, $J = 6.8$ Hz, 3H, alkyl $H^{K'}$), 0.53 (t, $J = 6.8$ Hz, 6H, alkyl H^K). ^{13}C NMR (125 MHz, DMSO): δ 160.51, 160.40, 158.47, 158.36, 157.29, 154.83, 153.60, 153.46, 146.87, 146.77, 146.51, 146.38, 139.65, 139.44, 134.45, 133.36, 130.65, 128.17, 128.00, 124.92, 124.73, 124.64, 124.56, 124.47, 119.04, 118.95, 89.51, 88.92, 75.88, 75.84, 31.91, 30.76, 30.68, 30.44, 29.87, 29.73, 29.58, 26.96, 26.75, 22.68, 13.74. ESI-MS (m/z): 1478.7 ($[M - 5PF_6]^{-5+}$; calcd m/z 1478.7), 1208.0 ($[M - 6PF_6]^{-6+}$; calcd m/z 1208.0), 1014.8 ($[M - 7PF_6]^{-7+}$; calcd m/z 1014.8), 869.8 ($[M - 8PF_6]^{-8+}$; calcd m/z 869.8), 757.1 ($[M - 9PF_6]^{-9+}$; calcd m/z 757.1), 666.8 ($[M - 10PF_6]^{-10+}$; calcd m/z 666.8), 593.1 ($[M - 11PF_6]^{-11+}$; calcd m/z 593.1), 592.6 ($[M - 12PF_6]^{-12+}$; calcd m/z 592.6).

Fe_9L_6 . Yield: 50%. 1H NMR (500 MHz, CD_3CN): δ 9.25 (s, 2H, tpy $H^{3'}$), 9.16 (s, 4H, tpy $H^{3'}$), 8.66 (d, $J = 8.0$ Hz, 2H, tpy $H^{3'}$), 8.54 (d, $J = 8.0$ Hz, 4H, tpy $H^{3'}$), 8.16 (s, 1H, Ph H^A), 8.14 (s, 2H, Ph H^A), 8.05–7.98 (m, 3H, tpy $H^{4'}$, Ph H^B), 7.94 (d, $J = 1.5$ Hz, 2H, Ph H^B), 7.85–7.72 (m, 7H, tpy H^4 , Ph $H^{C'}$, and Ph H^C), 7.27 (d, $J = 5.3$ Hz, 2H, tpy $H^{6'}$), 7.24–7.20 (m, 2H, tpy $H^{5'}$), 7.14 (d, $J = 5.7$ Hz, 4H, tpy H^6), 7.05–7.00 (m, 4H, tpy H^5), 4.43 (t, $J = 5.4$ Hz, 2H, alkyl $H^{D'}$), 4.34 (t, $J = 5.6$ Hz, 4H, alkyl H^D), 1.84 (m, 2H, alkyl $H^{E'}$), 1.80–1.73 (m, 4H, alkyl H^E), 1.48 (m, 2H, alkyl $H^{F'}$), 1.39 (m, 4H, alkyl H^F), 1.21 (m, 2H, alkyl $H^{G'}$), 1.13 (m, 4H, alkyl H^G), 0.96–0.79 (m, 18H, alkyl $H^{H'}$, alkyl $H^{H'}$, alkyl H^I , alkyl H^I , alkyl H^J , and alkyl H^J), 0.62 (t, $J = 6.9$ Hz, 3H, alkyl $H^{K'}$), 0.57 (t, $J = 6.9$ Hz, 6H, alkyl H^K). ^{13}C NMR (125 MHz, CD_3CN): δ 159.95, 159.84, 157.92, 157.81, 156.73, 154.27, 153.05, 152.91, 146.26, 145.84, 139.07, 138.88, 133.87, 132.96, 132.81, 130.12, 127.60, 127.44, 124.35, 124.17, 124.08, 123.99, 123.90, 122.87, 118.47, 118.38, 89.10, 88.34, 75.32, 75.25, 31.39, 31.34, 30.19, 30.12, 29.88, 29.30, 29.18, 29.12, 29.02, 26.37, 26.19, 22.11, 13.17. ESI-MS (m/z): 1884.5 ($[M - 6PF_6]^{-6+}$; calcd m/z 1884.5), 1594.7 ($[M - 7PF_6]^{-7+}$; calcd m/z 1594.7), 1377.2 ($[M - 8PF_6]^{-8+}$; calcd m/z 1377.2), 1208.1 ($[M - 9PF_6]^{-9+}$; calcd m/z 1208.1), 1072.7 ($[M - 10PF_6]^{-10+}$; calcd m/z 1072.7), 962.1 ($[M - 11PF_6]^{-11+}$; calcd m/z 962.1), 869.9 ($[M - 12PF_6]^{-12+}$; calcd m/z 869.9), 791.8 ($[M - 13PF_6]^{-13+}$; calcd m/z 791.8), 724.9 ($[M - 14PF_6]^{-14+}$; calcd m/z 724.9), 666.9 ($[M - 15PF_6]^{-15+}$; calcd m/z 666.9), 616.2 ($[M - 16PF_6]^{-16+}$; calcd m/z 616.2), 571.4 ($[M - 17PF_6]^{-17+}$; calcd m/z 571.4).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.inorgchem.7b02361.

Synthesis details for compounds **2–5**, NMR spectra (1H , ^{13}C , and ^{19}F 2D COSY), and ESI-MS isotope patterns for complexes Cd_6L_4 , Zn_6L_4 , Zn_9L_6 , Fe_6L_4 , and Fe_9L_6 (PDF)

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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